

Chapter 2: Atoms, Molecules, and Ions

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

CHM111
Lecture 2



Part I

Atomic Theory

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Ancient Atomism

Leucippus (~400 BC)

- ▶ Proposed that matter consists of tiny, indivisible particles called **atomos** (Greek: “uncuttable”).
 - ▶ Atoms differ in size, shape, and arrangement.
 - ▶ Empty space separates atoms.
-
- ▶ Aristotle rejected atomism, preferring “elements” that compose all matter: earth, water, air, fire.
 - ▶ Aristotle’s authority dominated for ~2000 years.

Ancient atomism was **philosophical**, not experimental.

It lacked the evidence needed to be a scientific theory, but it was on the right track.

ἄτομος [ˈa.tɔ.mɔs] *adjective* –
Ancient Greek
indivisible, uncuttable
 α (not) + $\tau\epsilon\mu\nu\omega$ (to cut)

Dalton's Atomic Theory (1807)

Five Postulates — Atomic Theory

1. Matter consists of extremely small, indivisible particles called **atoms**.
2. All atoms of a given element are **identical** in mass and properties.
3. Atoms of **different elements** have different properties.
4. Compounds contain atoms of two or more elements in **small, whole-number ratios**.
5. Atoms are **neither created nor destroyed** in a chemical reaction; they are rearranged.

Postulate 5 is the atomic-theory basis for the **Law of Conservation of Mass**.

Postulates 2 & 4 explain the **Law of Definite Proportions**.

Postulates 3 & 4 explain the **Law of Multiple Proportions**.



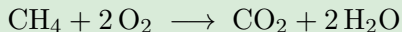
We now know Postulate 2 is not strictly true $\therefore$ **isotopes** exist.

Law of Conservation of Mass

Statement

The total mass of all substances present **does not change** during a chemical reaction.

Example: Combustion of Methane



► Mass of reactants ($\text{CH}_4 + 2 \text{O}_2$) = Mass of products ($\text{CO}_2 + 2 \text{H}_2\text{O}$)

- Atoms are merely **rearranged**, not created or destroyed.
- This law is the reason we balance chemical equations.

Law of Definite Proportions

Statement (Law of Constant Composition)

All pure samples of a given compound contain the same elements in the same **proportions by mass**, regardless of source.

Example: Water

- ▶ Water from a river, a glacier, or a lab all contain 11.19% H and 88.81% O by mass.
- ▶ The ratio is always $\text{H}:\text{O} = 1:8$ by mass.

Mass Fraction

- ▶ **Mass fraction** of element X =
$$\frac{\text{mass of X in sample}}{\text{total mass of sample}}$$
- ▶ Often expressed as a percentage (multiply by 100%).

Law of Multiple Proportions: An Introductory Example

Example: Collections of 5-lb (A) and 10-lb (B) Dumbbells

	Collection 1	Collection 2	Collection 3
Composition	1 A + 1 B	2 A + 1 B	4 A + 1 B
Mass of A	5 lb	10 lb	20 lb
Mass of B	10 lb	10 lb	10 lb
% A by mass	33%	50%	67%
% B by mass	67%	50%	33%
A:B mass ratio	$33/67 = 0.50$	$50/50 = 1.0$	$67/33 = 2.0$

Ratio of A:B mass ratios: Divide all mass ratios by the smallest mass ratio.

$$0.50/0.50 = 1.0; 1.0/0.50 = 2.0; 2.0/0.50 = 4.0$$

Since the mass-ratio calculation implies a fixed amount of B
(or thing in the denominator), we know the occurrence of A relative to B.

Col. 1: **AB** Col. 2: **A₂B** Col. 3: **A₄B**

Law of Multiple Proportions

When two elements form more than one compound, the masses of one element that combine with a fixed mass of the other are in a **ratio of small whole numbers**.

Example: Two Compounds of Elements B and R — (chem2e_ch2_1)

	Compound 1	Compound 2
% B by mass	42.8%	60.0%
% R by mass	57.2%	40.0%
B:R mass ratio	$42.8/57.2 = 0.748$	$60.0/40.0 = 1.50$

Ratio of B:R ratios: $\frac{1.50}{0.748} \approx \boxed{2.00} \Rightarrow$ a small whole number!

$\therefore$ Compound 1 is BR

Compound 2 is B₂R

Check Your Understanding

Nitrogen and oxygen form several compounds.

In compound A, 14.0 g of N combines with 16.0 g of O.

In compound B, 14.0 g of N combines with 32.0 g of O.

Which law does this illustrate, and what is the ratio?

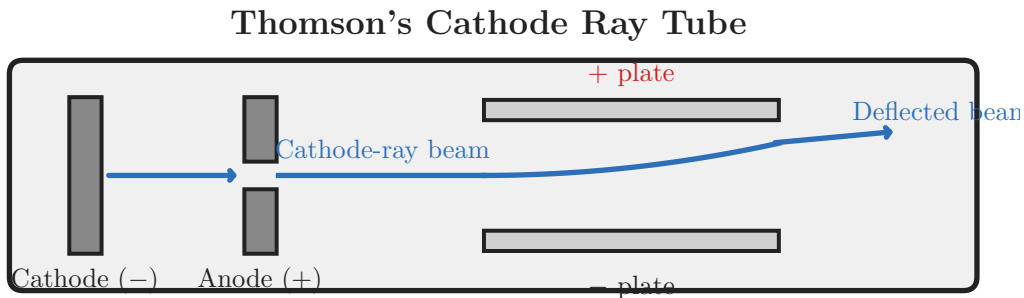
Come to lecture to see the answer.

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Discovery of the Electron: Thomson's Cathode Ray Tube



Discovery of the Electron: Thomson's Cathode Ray Tube

Thomson's Experiment (late 1800s)

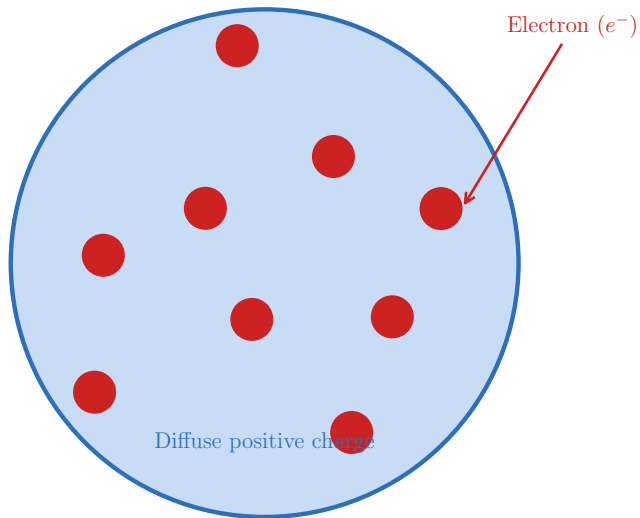
- ▶ Applied electric and magnetic fields to evacuated glass tubes.
- ▶ Observed a beam (“cathode rays”) deflected **toward the positive plate**.
- ▶ Concluded the beam consists of **negatively charged particles**.

What Was Discovered

- ▶ These particles are the same regardless of the cathode material.
- ▶ Their mass is ≈ 1800 times less than a hydrogen atom.
- ▶ These particles became known as **electrons**.

Thomson calculated the charge-to-mass ratio: $e/m = 1.759 \times 10^{11}$ C/kg.

Thomson's Plum Pudding Model



Thomson's Plum Pudding Model

Thomson's Atomic Model

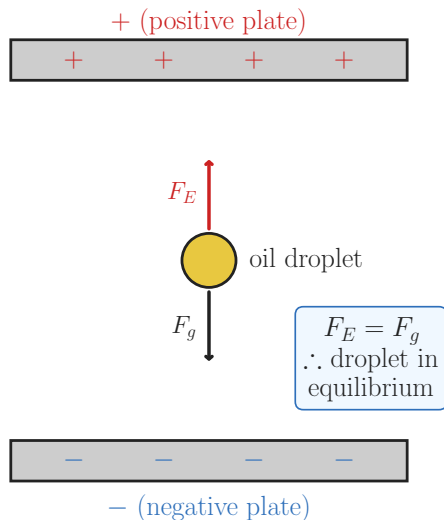
- ▶ Atoms are a **diffuse sphere of positive charge** with **electrons embedded** throughout.
- ▶ Overall charge = 0 (neutral atom).

This model is called the “plum pudding” model:

- ▶ Positive charge $\approx$ the pudding
- ▶ Electrons $\approx$ the fruit

This model was **disproved** by Rutherford's gold foil experiment a decade later.

Millikan's Oil Drop Experiment (1909)



Millikan's Oil Drop Experiment (1909)

What Millikan Measured

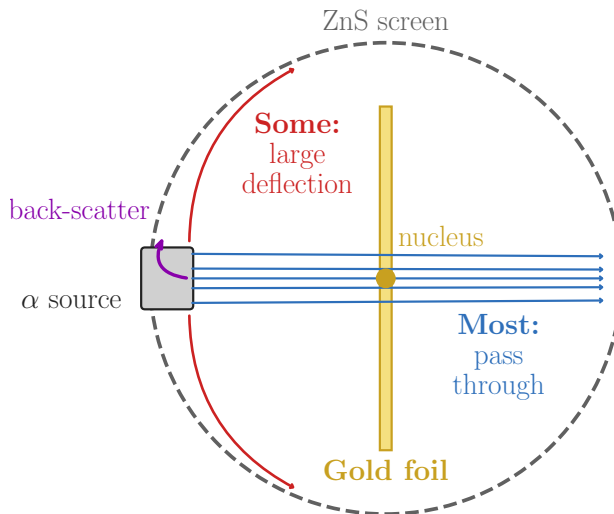
- ▶ Sprayed oil droplets between charged plates.
- ▶ Balanced gravitational and electric forces to hold droplets stationary.
- ▶ Measured the charge on each droplet.

Result

- ▶ All droplet charges were **multiples** of a single fundamental value.
- ▶ **Fundamental unit of charge:** $e = 1.602 \times 10^{-19} \text{ C}$
- ▶ Combined with Thomson's e/m ratio:

$$m_e = \frac{e}{e/m} = \frac{1.602 \times 10^{-19} \text{ C}}{1.759 \times 10^{11} \text{ C/kg}} = \boxed{9.107 \times 10^{-31} \text{ kg}}$$

Rutherford's Gold Foil Experiment (1909)



Rutherford's Gold Foil Experiment (1909)

Setup

- ▶ Aimed **alpha particles** (α , positively charged) at a thin gold foil.
- ▶ A circular screen detected where particles landed.

What Rutherford Expected

If Thomson's model were correct, alpha particles would pass straight through with only minor deflection.

What He Observed

- ▶ **Most:** passed straight through.
- ▶ **Some:** deflected at large angles.
- ▶ **A few:** bounced almost directly back.

Rutherford's Nuclear Model

Rutherford's Conclusion

Atoms contain a **tiny, dense, positively charged nucleus** surrounded by mostly **empty space** through which electrons move.

Key Features

- ▶ Nucleus contains essentially all of the atom's mass.
- ▶ Nucleus diameter: $\sim 10^{-15}$ m ($\sim 10^5$ times smaller than the atom).
- ▶ Electrons occupy the vast empty space outside the nucleus.

Rutherford's famous analogy:

“It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.”

Later work (Chadwick, 1932) identified **neutrons** — neutral particles in the nucleus with mass $\approx$ equal to protons.

Check Your Understanding

In Rutherford's gold foil experiment, what would have happened if Thomson's plum pudding model were correct?

Why was the actual result surprising?

Come to lecture to see the answer.

Part II

Atomic Structure

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

The Three Subatomic Particles

Particle	Symbol	Charge	Mass (amu)	Location
Proton	p^+	+1	1.00727	Nucleus
Neutron	n^0	0	1.00866	Nucleus
Electron	e^-	-1	0.000549	Outside nucleus

- ▶ 1 amu (atomic mass unit) $\equiv \frac{1}{12}$ the mass of a ^{12}C atom.
- ▶ The electron mass ($\approx 1/1836$ of a proton) is negligible in mass calculations.
- ▶ Protons + Neutrons $\approx$ all the atom's mass.

Atomic Number, Mass Number, and Neutrons

Atomic Number (Z)

Number of **protons** in the nucleus.

Defines the element: every carbon atom has $Z = 6$.

Mass Number (A)

Total: **protons + neutrons**:

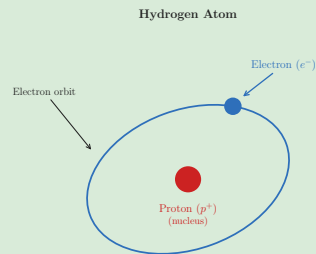
$$A = Z + N$$

where N = number of neutrons.

Neutrons (N)

$$N = A - Z$$

Hydrogen Atom



- ▶ $Z = 1$ (1 proton)
- ▶ $A = 1$ (1 proton, 0 neutrons)
- ▶ 1 electron (neutral atom)

Nuclide Symbols

Standard Notation

$${}^A_Z\text{X} \quad \text{or equivalently} \quad \text{X-}A$$

where X = chemical symbol, A = mass number, Z = atomic number.

Examples

- ▶ ${}^1_1\text{H}$ (or H-1): 1 proton, 0 neutrons
- ▶ ${}^4_2\text{He}$ (or He-4): 2 protons, 2 neutrons
- ▶ ${}^{12}_6\text{C}$ (or C-12): 6 protons, 6 neutrons
- ▶ ${}^{238}_{92}\text{U}$ (or U-238): 92 protons, 146 neutrons

Z can be omitted in writing because the element symbol already specifies it.

A is always given explicitly because it varies between isotopes.

Isotopes

Definition

Isotopes are atoms of the *same* element (same Z) that differ in the number of neutrons (different A).

Isotopes of Carbon



6 protons

6 neutrons

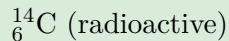
98.89% abundance



6 protons

7 neutrons

1.11% abundance



6 protons

8 neutrons

trace abundance

- ▶ Isotopes are **chemically identical** — same number of electrons, same bonding behavior.
- ▶ Isotopes have different **masses** and may differ in radioactivity.
- ▶ ${}^{14}\text{C}$ is used in **radiocarbon dating** of archaeological artifacts.

Ions: Cations and Anions

Ion

An atom (or molecule) with an **unequal number of protons and electrons** — it carries a net charge.

Cation (positive ion)

- ▶ **Loses** one or more electrons.
- ▶ More protons than electrons.
- ▶ *Examples:* Na^+ , Mg^{2+} , Al^{3+}

Anion (negative ion)

- ▶ **Gains** one or more electrons.
- ▶ More electrons than protons.
- ▶ *Examples:* Cl^- , O^{2-} , N^{3-}

Neutral atom: $\# \text{electrons} = \# \text{protons}$

If $\# e^- \neq \# p^+$, we have an ion.

e.g. A sodium atom ($11 p^+$, $11 e^-$) loses one electron $\Rightarrow \text{Na}^+$ ($11 p^+$, $10 e^-$).

Check Your Understanding

An atom of ${}_{17}^{35}\text{Cl}$ gains one electron to form a chloride ion.

How many protons, neutrons, and electrons does the ion have?

What is its charge?

Come to lecture to see the answer.

Average Atomic Mass

Why Average?

Most elements exist as a **mixture of isotopes**. The atomic mass listed on the periodic table is a **weighted average** of all natural isotopes.

Formula

$$\bar{m} = \sum_i (\text{fractional abundance}_i \times m_i)$$

- ▶ Fractional abundance = (percent abundance) / 100
- ▶ m_i = isotopic mass in amu

Isotopic abundances and masses are measured by **mass spectrometry**.

Example: Average Atomic Mass of Boron

Average Atomic Mass of Boron — (chem2e_ch2_ex2.4)

Boron has two stable isotopes:

^{10}B : 19.9% abundance, mass = 10.0129 amu

^{11}B : 80.1% abundance, mass = 11.0093 amu

Calculate the average atomic mass of boron.

$$\bar{m} = (0.199)(10.0129 \text{ amu}) + (0.801)(11.0093 \text{ amu})$$

$$\bar{m} = 1.993 \text{ amu} + 8.818 \text{ amu}$$

$$\bar{m} = 10.811 \text{ amu}$$

The average is **closer to 11** because ^{11}B is more abundant ($\sim 80\%$).

Compare with the periodic table value: 10.81 amu.

Check Your Understanding

Chlorine has two stable isotopes:

^{35}Cl : 75.77% abundance, mass = 34.969 amu

^{37}Cl : 24.23% abundance, mass = 36.966 amu

Without calculating, is the average atomic mass of chlorine closer to 35 or 37?

Come to lecture to see the answer.

Part III

Molecules and the Periodic Table

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Types of Chemical Formulas

Molecular Formula

Shows the **actual number** of each type of atom in one molecule.

Examples:

H_2O , CO_2 , $\text{C}_6\text{H}_{12}\text{O}_6$

A **molecular formula** is always a whole-number multiple of the **empirical formula**.

e.g. Glucose: molecular = $\text{C}_6\text{H}_{12}\text{O}_6$; empirical = CH_2O (ratio 6:12:6 = 1:2:1).

Empirical Formula

Shows the **simplest whole-number ratio** of atoms.

Examples:

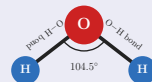
H_2O , CO_2 , CH_2O (for glucose)

Structural Formula

Shows how atoms are **connected** (bonded) to each other.

Structural Formula of Water

Molecular formula: H_2O



Reveals molecular geometry.

Subscripts vs. Coefficients

Important Distinction

- ▶ A **subscript** tells how many atoms of that element are in *one* molecule.
- ▶ A **coefficient** tells how many molecules (or formula units) are present.

Examples

- | | |
|--|---|
| ▶ H_2 = one molecule with 2 H atoms bonded together | ▶ H_2O = 2 H + 1 O bonded in one molecule |
| ▶ 2H = two separate H atoms (not bonded) | ▶ $3\text{H}_2\text{O}$ = 3 water molecules (6 H + 3 O total) |
| ▶ 2H_2 = two molecules of H_2 (4 H atoms total) | |

When writing chemical equations, you change **coefficients** to balance — never subscripts (that would change the substance itself).

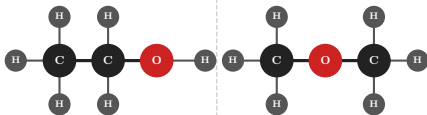
Isomers

Definition

Isomers are compounds that share the same **molecular formula** but have **different structural arrangements**, and therefore different properties.

Isomers of C_2H_6O

Both: molecular formula C_2H_6O

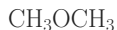


Ethanol



bp = 78.4 °C

Dimethyl ether



bp = -24.8 °C

Structure matters.

Ethanol is drinking alcohol, found in liquor.

Dimethyl ether is primarily used as an aerosol propellant, a chemical manufacturing feedstock, a clean-burning alternative fuel, and an active ingredient in medical or industrial cold sprays.

Molecular Mass (Formula Mass)

Definition

The **molecular mass** (also called formula mass) is the sum of the **atomic masses** of all atoms in one molecule, expressed in amu.

Examples

$$M(\text{H}_2\text{O}) = 2(1.008) + 15.999 = \boxed{18.02 \text{ amu}}$$

$$M(\text{CO}_2) = 12.011 + 2(15.999) = \boxed{44.01 \text{ amu}}$$

$$M(\text{CO}) = 12.011 + 15.999 = \boxed{28.01 \text{ amu}}$$

Note that CO_2 and CO have very different molecular masses despite containing the same elements — because they differ in the **number** of oxygen atoms.

Check Your Understanding

The molecular formula for glucose is $\text{C}_6\text{H}_{12}\text{O}_6$.

- (a) What is its empirical formula?
- (b) How many total atoms are in one glucose molecule?

Come to lecture to see the answer.

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

1

1.00794

H

Hydrogen

3

6.941

Li

Lithium

11

22.98977

Na

Sodium

19

39.0983

K

Potassium

37

85.4678

Rb

Rubidium

55

132.90545

Cs

Cesium

87

223

Fr

Francium

2

I IA

4

9.012182

Be

Beryllium

12

24.305

Mg

Magnesium

20

40.078

Ca

Calcium

38

87.62

Sr

Strontium

56

137.327

Ba

Barium

88

226

Ra

Radium

13

10.811

B

Boron

14

12.0107

C

Carbon

15

14.0067

N

Nitrogen

16

15.9994

O

Oxygen

17

18.9984032

F

Fluorine

18

20.1797

Ne

Neon

13

26.982

Al

Aluminium

14

28.0855

Si

Silicon

15

30.973761

P

Phosphorus

16

32.065

S

Sulphur

17

35.453

Cl

Chlorine

18

39.948

Ar

Argon

3

IIA

4

IVB

5

VB

6

VIB

7

VII B

8

VIII B

9

VIII B

10

VIII B

11

IB

12

IIB

13

IIIA

14

IVA

15

VA

16

VIA

17

VIIA

18

VIIIA

21

44.95591

Sc

Scandium

22

47.867

Ti

Titanium

23

50.9415

V

Vanadium

24

51.9961

Cr

Chromium

25

54.938049

Mn

Manganese

26

55.845

Fe

Iron

27

58.9332

Co

Cobalt

28

58.6934

Ni

Nickel

29

63.546

Cu

Copper

30

65.409

Zn

Zinc

31

69.723

Ga

Gallium

32

72.64

Ge

Germanium

33

74.9216

As

Arsenic

34

78.96

Se

Selenium

35

79.904

Br

Bromine

36

83.798

Kr

Krypton

39

88.90585

Y

Yttrium

40

91.224

Zr

Zirconium

41

92.90638

Nb

Niobium

42

95.94

Mo

Molybdenum

43

98

Tc

Technetium

44

101.07

Ru

Ruthenium

45

102.9055

Rh

Rhodium

46

106.42

Pd

Palladium

47

107.8682

Ag

Silver

48

112.411

Cd

Cadmium

49

114.82

In

Indium

50

118.71

Sn

Tin

51

121.76

Sb

Antimony

52

127.6

Te

Tellurium

53

126.90447

I

Iodine

54

131.293

Xe

Xenon

57-71

La-Lu

72

178.49

Hf

Hafnium

73

180.9479

Ta

Tantalum

74

183.84

W

Tungsten

75

186.207

Re

Rhenium

76

190.23

Os

Osmium

77

192.217

Ir

Iridium

78

195.078

Pt

Platinum

79

196.96655

Au

Gold

80

200.59

Hg

Mercury

81

204.38

Tl

Thallium

82

207.2

Pb

Lead

83

208.98038

Bi

Bismuth

84

209

Po

Polonium

85

210

At

Astatine

86

222

Rn

Radon

89-103

Ac-Lr

104

261

Rf

Rutherfordium

105

262

Db

Dubnium

106

266

Sg

Seaborgium

107

264

Bh

Bohrium

108

277

Hs

Hassium

109

268

Mt

Meitnerium

110

281

Ds

Darmstadtium

111

272

Rg

Roentgenium

112

285

Cn

Copernicium

113

284

Nh

Nihonium

114

289

Fl

Flerovium

115

289

Mc

Moscovium

116

293

Lv

Livermorium

117

294

Ts

Tennesse

118

294

Og

Ognesson

57

138.91

La

Lanthanum

58

140.116

Ce

Cerium

59

140.90765

Pr

Praseodymium

60

144.24

Nd

Neodymium

61

145

Pm

Promethium

62

150.36

Sm

Samarium

63

151.964

Eu

Europium

64

157.25

Gd

Gadolinium

65

158.92534

Tb

Terbium

66

162.5

Dy

Dysprosium

67

164.93032

Ho

Holmium

68

167.259

Er

Erbium

69

168.93421

Tm

Thulium

70

173.04

Yb

Ytterbium

71

174.967

Lu

Lutetium

89

227

Ac

Actinium

90

232.0381

Th

Thorium

91

231.03588

Pa

Protactinium

92

238.02891

U

Uranium

93

237

Np

Neptunium

94

244

Pu

Plutonium

95

243

Am

Americium

96

247

Cm

Curium

97

247

Bk

Berkelium

98

251

Cf

Californium

99

252

Es

Einsteinium

100

257

Fm

Fermium

101

258

Md

Mendelevium

102

259

No

Nobelium

103

262

Lr

Lawrencium

Z

mass

Symbol

name

black: natural

gray: man-made

Alkali Metal

Alkaline Earth Metal

Metal

Metalloid

Non-metal

Halogen

Noble Gas

Lanthanide/Actinide

Periodic Table of Elements

T. K. Deskins

Atomic-mass values from the 2020 Atomic Mass Evaluation

Organization of the Periodic Table

Periodic Law

The properties of the elements are periodic functions of their atomic numbers.

Periods (rows)

- ▶ 7 horizontal rows (periods).
- ▶ Elements ordered by **increasing atomic number** left to right.
- ▶ Properties change systematically across a period.

Groups (columns)

- ▶ 18 vertical columns (groups).
- ▶ Elements in the same group have **similar chemical properties**.
- ▶ Groups 1, 2, and 13–18 are the **main groups**.
- ▶ Groups 3–12 are the **transition metals**.

Metals, Nonmetals, and Metalloids

Metals

- ▶ Shiny, malleable, ductile
- ▶ Good conductors of heat and electricity
- ▶ **Lose** electrons to form cations
- ▶ Left and center of periodic table

Nonmetals

- ▶ Dull, brittle (if solid)
- ▶ Poor conductors
- ▶ **Gain** electrons to form anions (or share electrons)
- ▶ Upper right of periodic table

Metalloids

- ▶ Intermediate properties
- ▶ Moderate conductivity (**semiconductors**)
- ▶ *Examples:* Si, Ge, As, Sb, Te
- ▶ Staircase boundary between metals and nonmetals

Transition metals (Groups 3–12) readily form cations of **varying charge**.
Main-group metals typically form cations of a **fixed charge**.

Named Element Families

Group	Name	Key Property	Examples
1	Alkali metals	Very reactive; form 1+ ions	Li, Na, K
2	Alkaline earth metals	Reactive; form 2+ ions	Mg, Ca, Ba
15	Pnictogens	Nitrogen group	N, P, As
16	Chalcogens	Oxygen group; form 2− ions	O, S, Se
17	Halogens	Highly reactive; form 1− ions	F, Cl, Br, I
18	Noble gases	Very unreactive (full shell)	He, Ne, Ar

Elements in the same group have the same number of **valence electrons**, which determines their chemistry.

Check Your Understanding

Without looking at a periodic table, classify each element as a **metal**, **nonmetal**, or **metalloid**, and identify its group family if it has a named one:

- (a) Potassium (K, $Z = 19$)
- (b) Fluorine (F, $Z = 9$)
- (c) Silicon (Si, $Z = 14$)
- (d) Argon (Ar, $Z = 18$)

Come to lecture to see the answer.

Part IV

Compounds and Nomenclature

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Ion Formation

During chemical reactions, **electrons transfer** from one atom to another, producing electrically charged particles (ions). This is how ions are formed.

Metals → Cations

- ▶ Metal Atoms **lose** electrons to form cations.
- ▶ Group 1 metals: lose 1 $e^- \Rightarrow 1+$ charge ($\text{Na} \rightarrow \text{Na}^+$)
- ▶ Group 2 metals: lose 2 $e^- \Rightarrow 2+$ charge ($\text{Mg} \rightarrow \text{Mg}^{2+}$)
- ▶ Group 13 metals: lose 3 $e^- \Rightarrow 3+$ charge ($\text{Al} \rightarrow \text{Al}^{3+}$)

Nonmetal Atoms → Anions

- ▶ Nonmetals **gain** electrons to form anions.
- ▶ Group 17 (halogens): gain 1 $e^- \Rightarrow 1-$ charge ($\text{Cl} \rightarrow \text{Cl}^-$)
- ▶ Group 16 (chalcogens): gain 2 $e^- \Rightarrow 2-$ charge ($\text{O} \rightarrow \text{O}^{2-}$)
- ▶ Group 15: gain 3 $e^- \Rightarrow 3-$ charge ($\text{N} \rightarrow \text{N}^{3-}$)

Common Monatomic Ions

Common Cations

Na^+	sodium	Fe^{2+}	iron(II)
K^+	potassium	Fe^{3+}	iron(III)
Mg^{2+}	magnesium	Cu^+	copper(I)
Ca^{2+}	calcium	Cu^{2+}	copper(II)
Ba^{2+}	barium	Zn^{2+}	zinc
Al^{3+}	aluminum	NH_4^+	ammonium

Common Anions

F^-	fluoride	O^{2-}	oxide
Cl^-	chloride	S^{2-}	sulfide
Br^-	bromide	N^{3-}	nitride
I^-	iodide	P^{3-}	phosphide

Monatomic anion names end in **-ide**.

Polyatomic Ions

Definition

A **polyatomic ion** is a charged group of two or more bonded atoms that acts as a single unit.

Formula	Name	Charge	Formula	Name	Charge
NH_4^+	ammonium	1+	OH^-	hydroxide	1-
NO_3^-	nitrate	1-	NO_2^-	nitrite	1-
CO_3^{2-}	carbonate	2-	HCO_3^-	bicarbonate	1-
SO_4^{2-}	sulfate	2-	SO_3^{2-}	sulfite	2-
PO_4^{3-}	phosphate	3-	CN^-	cyanide	1-

These polyatomic ions (and others) must be memorized.

Ionic Compounds

Definition

An **ionic compound** consists of cations and anions arranged in a crystal lattice, held together by **electrostatic attraction** (ionic bonds).

Writing Ionic Formulas

- ▶ The overall compound must be **electrically neutral**.
- ▶ Cation charge + anion charge = 0.
- ▶ *Example:* Mg^{2+} and F^{-}
Need 2 F^{-} per Mg^{2+} : **MgF_2**
- ▶ *Example:* Mg^{2+} and OH^{-}
Need 2 OH^{-} per Mg^{2+} : **$\text{Mg}(\text{OH})_2$**

Ionic compounds:

- ▶ Typically have **high melting points**.
- ▶ Conduct electricity when **molten** or **dissolved in water**.
- ▶ Often form crystalline solids.

Molecular (Covalent) Compounds

Definition

A **molecular compound** (covalent compound) consists of discrete molecules formed when nonmetals **share electrons** via covalent bonds.

Examples

- ▶ H_2O (water)
- ▶ CO_2 (carbon dioxide)
- ▶ NH_3 (ammonia)
- ▶ CH_4 (methane)
- ▶ N_2O_4 (dinitrogen tetroxide)

Molecular

Nonmetal + nonmetal
 Share electrons
 Discrete molecules
 Lower melting pt

Ionic

Metal + nonmetal
 Transfer electrons
 Crystal lattice
 Higher melting pt

Check Your Understanding

Classify each as **ionic** or **molecular**:

- (a) NaCl (table salt)
- (b) H₂O (water)
- (c) MgO (magnesium oxide)
- (d) SO₂ (sulfur dioxide)

Come to lecture to see the answer.

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Naming Ionic Compounds: Monatomic Ions

Rule

Name the **cation** first, then the **anion** with an **-ide** suffix.

Examples

- ▶ NaCl = sodium chloride
- ▶ KBr = potassium bromide
- ▶ MgO = magnesium oxide
- ▶ CaF_2 = calcium fluoride
- ▶ Al_2O_3 = aluminum oxide
- ▶ MgF_2 = magnesium fluoride

- ▶ Do **not** include subscripts in the name.
- ▶ Do **not** say “magnesium di-fluoride” — the charge ratio is already fixed by the ion charges.

Naming Ionic Compounds: Variable-Charge Metals

Rule

When a metal can form more than one cation charge, indicate the charge with a **Roman numeral** in parentheses after the metal name.

Examples

- ▶ FeCl_2 = **iron(II) chloride** (Fe^{2+})
- ▶ FeCl_3 = **iron(III) chloride** (Fe^{3+})
- ▶ CuO = **copper(II) oxide** (Cu^{2+})
- ▶ Cu_2O = **copper(I) oxide** (Cu^{+})
- ▶ SnCl_4 = **tin(IV) chloride** (Sn^{4+})

Metals with Variable Charge

- ▶ Fe (iron): 2+, 3+
- ▶ Cu (copper): 1+, 2+
- ▶ Sn (tin): 2+, 4+
- ▶ Pb (lead): 2+, 4+
- ▶ Hg (mercury): 1+, 2+
- ▶ Co, Cr, Mn, Ni ...

Naming Ionic Compounds: Polyatomic Ions

Rule

Use the **name of the polyatomic ion** directly; do not add -ide unless the ion is CN^- (cyanide) or OH^- (hydroxide).

Examples

- ▶ NaOH = sodium hydroxide
- ▶ $\text{Al}_2(\text{SO}_4)_3$ = aluminum sulfate
- ▶ $\text{Mg}(\text{NO}_3)_2$ = magnesium nitrate
- ▶ $(\text{NH}_4)_2\text{CO}_3$ = ammonium carbonate
- ▶ $\text{Ca}_3(\text{PO}_4)_2$ = calcium phosphate
- ▶ $\text{Mg}(\text{OH})_2$ = magnesium hydroxide

Note parentheses: use them around a polyatomic ion when the subscript > 1 .
e.g. $\text{Mg}(\text{OH})_2$ — not MgOH_2 (which is chemically ambiguous).

Naming Ionic Hydrates

Definition

A **hydrate** is an ionic compound with a specific number of water molecules incorporated into its crystal structure.

Naming Rule

Name the ionic compound normally, then add a Greek prefix + **hydrate**.

Examples

- ▶ $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ = **copper(II) sulfate pentahydrate**
- ▶ $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ = **calcium chloride dihydrate**
- ▶ $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ = **sodium sulfate decahydrate**

The “ $\cdot$ ” (centered dot) in the formula separates the ionic compound from the water molecules.

Check Your Understanding

Name each compound:

- (a) K_2S
- (b) Fe_2O_3
- (c) $\text{Mg}(\text{HCO}_3)_2$
- (d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Come to lecture to see the answer.

Naming Molecular Compounds: Greek Prefixes

Rule

Use **Greek prefixes** to indicate the number of each type of atom. Name the more metallic element first; second element gets an **-ide** suffix. Drop the trailing vowel of the prefix before a vowel (e.g. monooxide → monoxide).

Prefixes

1	mono-	6	hexa-
2	di-	7	hepta-
3	tri-	8	octa-
4	tetra-	9	nona-
5	penta-	10	deca-

Examples

- ▶ CO_2 = carbon dioxide
- ▶ CO = carbon monoxide
- ▶ SO_2 = sulfur dioxide
- ▶ N_2O_4 = dinitrogen tetroxide
- ▶ NO_2 = nitrogen dioxide
- ▶ PCl_5 = phosphorus pentachloride
- ▶ SF_6 = sulfur hexafluoride

Naming Binary Acids

A **binary acid** consists of hydrogen bonded to one nonmetal element, dissolved in water.

Naming Rule

1. Replace “hydrogen” with the prefix **hydro-**.
2. Change the nonmetal ending to **-ic**.
3. Add the word **acid**.

Examples

- ▶ HCl (pure gas): hydrogen chloride
HCl(aq): **hydrochloric acid**
- ▶ HBr(aq): **hydrobromic acid**
- ▶ HF(aq): **hydrofluoric acid**
- ▶ H₂S(aq): **hydrosulfuric acid**

Naming Oxyacids

Definition

An **oxyacid** contains hydrogen, oxygen, and at least one other element.

Naming Rule

Start from the name of the **oxyanion**, then:

- ▶ If anion ends in **-ate** $\Rightarrow$ acid ends in **-ic acid**
- ▶ If anion ends in **-ite** $\Rightarrow$ acid ends in **-ous acid**

- ▶ NO_3^- = nitrate $\Rightarrow$ HNO_3 = **nitric acid**
- ▶ NO_2^- = nitrite $\Rightarrow$ HNO_2 = **nitrous acid**
- ▶ SO_4^{2-} = sulfate $\Rightarrow$ H_2SO_4 = **sulfuric acid**

- ▶ SO_3^{2-} = sulfite $\Rightarrow$ H_2SO_3 = **sulfurous acid**
- ▶ CO_3^{2-} = carbonate $\Rightarrow$ H_2CO_3 = **carbonic acid**
- ▶ PO_4^{3-} = phosphate $\Rightarrow$ H_3PO_4 = **phosphoric acid**

Check Your Understanding

Give the name or formula as appropriate:

- (a) N_2O_5 (molecular compound)
- (b) **sulfur tetrafluoride**
- (c) H_2SO_4
- (d) **hydroiodic acid**

Come to lecture to see the answer.

Check Your Understanding

Write the formula for each compound:

- (a) calcium phosphate
- (b) iron(III) nitrate
- (c) ammonium sulfate
- (d) dinitrogen monoxide

Come to lecture to see the answer.

Summary: Chapter 2

- ▶ **Dalton's atomic theory** (1807): atoms are small, indivisible, and rearranged (not destroyed) in reactions. Explains Laws of Conservation of Mass, Definite Proportions, and Multiple Proportions.
- ▶ **Subatomic particles**: protons (+, in nucleus), neutrons (0, in nucleus), electrons (−, outside nucleus).
- ▶ **Atomic number** Z = # protons; **mass number** A = protons + neutrons; **isotopes** have same Z , different A .
- ▶ **Average atomic mass** = weighted average over natural isotopes.
- ▶ **Molecular formula** = actual atom count; **empirical formula** = simplest ratio.
- ▶ The **periodic table** organizes elements by increasing Z into periods (rows) and groups (columns) of similar properties.
- ▶ **Ionic compounds**: cation + anion; electrically neutral crystal lattice.
- ▶ **Molecular compounds**: nonmetals sharing electrons.